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ORIGINAL ARTICLE

Improved ZND model for solving dynamic linear complex matrix equation and its application

Zhiyuan Song[1] · Zhenyao Lu[1] · Jiahao Wu[1] · Xiuchun Xiao[1] · Guancheng Wang[1]

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Abstract

The online solving of a dynamic linear complex matrix equation (DLCME) is commonly encountered in many fields, and it exists for lots of engineering applications. For solving the DLCME, the gradient neural dynamics (GND) and the zeroing neural dynamics (ZND) are usually designed based on a predefined error function in scalar or matrix form to monitor the residual error convergence, respectively. In this paper, in order to improve the adaptability as the residual error decreasing with time and to release the limitation of convex activation function, an improved zeroing neural dynamics (IZND) model is proposed with a residual-based adaptive coefficient and a non-convex activation function. By using the Lyapunov stability theory, the global convergence and noise suppression of the proposed IZND model are theoretically discussed under the noise-free and noise-perturbed circumstances. Then, simulative verifications and a dynamic acoustic source localization application illustrate the efficacy and superiority of the proposed model for solving the DLCME. Comparing with some existing GND and ZND models, the proposed IZND model shows advanced performance in solution accuracy, noise suppression and convergence rate.

Keywords Dynamic problems · Linear complex matrix equation · Zeroing neural dynamics (ZND) · Adaptive coefficient · Non-convex activation function

1 Introduction

The online solving of a dynamic linear complex matrix equation (DLCME) is one of the essential problems in various scientific and engineering fields, such as robot kinematics [1–4], mobile object localization [5, 6], and acoustic source localization [7]. In the past decades, many kinds of neural dynamic methods have emerged with the intensive study of machine learning theory. Among these methods, one method called recurrent neural dynamic (RND) has been applied to solve various problems [8, 9]. Subsequently, some scholars have improved on recurrent neural dynamic (RND) by proposing gradient-based neural dynamics (GND). The traditional GND is intrinsically devised for solving the time-invariant issues. As described by Zhang and Chen et al. [10, 11], this kind of method usually suffers from significant lagging errors when it is applied to dynamic issues. Liao et al. proposed a modified gradient-based neural dynamic (MGND) model [12] that breaks the limitation of the traditional GND model in addressing dynamic issues. Moreover, there is another method called zeroing neural dynamics or Zhang neural dynamics (ZND) that can also be well-used to address the dynamic issues [13, 14], which has achieved great success to improve the performance of global convergence and solution accuracy [15–17].

Nevertheless, in the last decade, some researchers have conducted some studies of the problems in the complex-valued fields. Song et al. [18] proposed a complex-valued

✉ Xiuchun Xiao
xiaoxc@gdou.edu.cn

Zhiyuan Song
yunshangzhuixi@gmail.com

Zhenyao Lu
jdshxn@sina.com

Jiahao Wu
gdouwjh@gmail.com

Guancheng Wang
wanggc@gdou.edu.cn

[1] College of Electronic and Information Engineering, Guangdong Ocean University, Zhanjiang 524088, China

gradient neural dynamic (CVGND) for solving inverse and pseudo-inverse of a complex matrix. However, the CVGND has many limitations, e.g., its activation function must be a convex and monotonically increasing odd function. Thus, Zhang et al. continued this research and recommended a complex-valued zeroing neural dynamic (CVZND) for solving complex-valued field issues [19]. It simultaneously releases the limitation of convex activation function while holding the characteristics of global and online convergence to the theoretical solution. Anteriorly, Zhang et al. designed the study of stationary quadratic programming issues in complex-valued fields according to the same design mechanism [20]. After that, Xiao recommended a finite-time convergent neural dynamics in [21] by applying sufficient time diffraction data of time-varying complex-valued matrix parameters. Beyond that, Xiao et al. proposed a model named PC-CVZND that can be used to solve time-varying complex-valued problems by constructing a new time-varying coefficient function [22]. The PC-CVZND model can obtain better results than the fixed coefficients and the existing coefficient variation function employed in the CVZND because of the acceleration effect of the new coefficient variation function. More striking researches were with the AFT-ZND model suggested by Jia et al. in [23], and it integrated an adaptive fuzzy control strategy and then adaptively adjusted its convergence rate according to the computational error value.

However, in this class, most of the above methods are restricted by the convex activation function. Xiao et al. defined a method for building a particular kind of activation functions allowed saturation that improve the convex restraint of the ZND model [24]. Meanwhile, Xiao et al. proposed some methods to help determine whether a ZND model activated by a predefined activation function has finite time convergence and predefined time convergence [25]. Based on them, Jiang et al. designed and theorized a non-convex constrained zeroing neural dynamic (NCZND) model in [26], which constructed several new non-convex and bounded constrained complex-valued activation functions based on through continuing the study on the foundation of real-valued activation functions. In addition, it is known that the robustness is one of the important merit of any algorithm [27, 28]. Liao et al. proposed a unified design formulation of the ZND model for facilitating finite-time convergence of the model while resolving noise disturbances [29]. Subsequently, Liao continued his research on the theory by promoting a prescribed-time convergent and noise-tolerant ZND (PTCNTZND) model for calculating real-time-dependent quadratic programming under noisy environments[30]. Prior to this, a noise-tolerant zeroing neural dynamics (NTZND) model that can resist noise interference was also proposed for solving the non-stationary Lyapunov equation by Yan and Xiao [31]. Moreover, Chen et al. proposed an optimization algorithm capable of being used for computed tomography diagnosis, extended a novel hyper-twisted nulling neural network applied to a mobile robot manipulator, and optimized a model capable of being used for parallel robot tracking control, all of which are applications based on neural dynamic methods [32–34].

Inspired by the above discussion, we propose a non-convex activation and residual-based adaptive coefficient zeroing neural dynamic (IZND) model in combination with the experience of the above algorithms. It is worth mentioning that this adaptive coefficient is a residual-based segmentation function with considerable flexibility to improve the adaptability of residuals decreasing with time. Furthermore, under the same conditions, the convergence accuracy of the IZND model with our adaptive coefficients is better than that of the ZND model with either of the time-varying coefficients in [22]. What's more important, this adaptive factor improves the performance of the ZND model against noise perturbations. Apart from this, we relax the convex restriction on the activation function by applying a non-convex activation function, and this non-convex activation function also serves to suppress the noise to some extent, improving the robustness of the model.

The remainder of this paper is laid out as follows: In Sect. 2, the conventional GND and ZND models for solving the DLCME are described as related preliminaries for later comparisons purpose. In Sect. 3, the construction of non-convex activation function and the design of residual-based adaptive coefficient are described in detail; then, the IZND model is proposed. In sect. 4, the global convergence of the IZND model is demonstrated as well as its robustness under constant and dynamic noise. In Sect. 5, the superiority for the presented IZND model is verified in simulations, especially in comparison with other dynamic coefficient models. In Sect. 6, the paper is summarized. The major contributions of the paper are listed as follows.

(1) In this paper, a IZND model with non-convex activation and residual-based adaptive coefficient is designed for solving the dynamic linear complex matrix equations online. The IZND model has better convergence rate and accuracy, distinguishing it from the general fixed coefficient zeroing neural dynamics and other dynamic parametric zeroing neural dynamics.
(2) The global convergence of the IZND model is proved analytically through the Lyapunov stability theory. Meanwhile, the robustness of the IZND model in solving the DLCME accompanied with constant and dynamic noise perturbations is

demonstrated using a rigorous and well thought-out logic, respectively.

(3) Through simulation experiments, the effectiveness of the IZND model in solving the DLCME is verified, and the advantages of the IZND model in terms of convergence accuracy and convergence speed compared with general gradient-based neural dynamics and zeroing neural dynamics are demonstrated.

2 Problem formulation and preliminary schemes

This section mainly discusses the problem formulation, and for comparative purpose, two preliminary schemes, i.e., the traditional GND and ZND schemes, are introduced for solving the DLCME.

2.1 Problem formulation

Generally, a typical DLCME can be formulated as follows:

$$\mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) = \mathbf{I}, \tag{1}$$

where $t \in [0, +\infty)$ denotes time, $\mathbf{I} \in \mathbb{C}^{n\times n}$ represents the identity matrix, and $\mathbf{A}(t) \in \mathbb{C}^{n\times n}$ and $\mathbf{B}(t) \in \mathbb{C}^{n\times n}$ are assumed as smoothly dynamic complex matrices. The aim of the solution scheme is to obtain the uncertain matrix $\mathbf{X}(t) \in \mathbb{C}^{n\times n}$ that satisfies Eq. (1).

2.2 Preliminary schemes

2.2.1 Traditional GND model

Inspired by the method of constructing gradient-based neural dynamic in [35], an energy function $\mathbf{s} = \|\mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) - \mathbf{I}\|_{\mathrm{F}}^2/2$ was designed by using the Frobenius norm and then obtained the following GND model similar to the gradient descent method.

$$\begin{aligned}\dot{\mathbf{X}}(t) &= -\beta\frac{\partial \mathbf{s}}{\partial \mathbf{X}}\\ &= -\beta\mathbf{A}^{\mathrm{T}}(t)(\mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) - \mathbf{I})\mathbf{B}^{\mathrm{T}}(t),\end{aligned} \tag{2}$$

in which $\beta > 0$ denotes a design parameter related to convergence performance.

2.2.2 Traditional ZND model

According to the design idea of the traditional ZND model, an error function can be defined for solving the above DLCME as

$$\mathbf{E}(t) = \mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) - \mathbf{I}. \tag{3}$$

To force the residual error $\mathbf{E}(t) \in \mathbb{C}^{n\times n}$ to converge to zero, the traditional ZND model can be designed as follows:

$$\dot{\mathbf{E}}(t) = -\gamma\Psi(\mathbf{E}(t)), \tag{4}$$

where $\Psi(\cdot): \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ represents a complex-valued activation function and $\gamma > 0$ denotes a design parameter related to the convergence performance. Moreover, substituting Eq. (3) into Eq. (4) easily has

$$\begin{aligned}\dot{\mathbf{X}}(t) = &-\mathbf{A}^{-1}(t)(\dot{\mathbf{A}}(t)\mathbf{X}(t)\mathbf{B}(t) + \mathbf{A}(t)\mathbf{X}(t)\dot{\mathbf{B}}(t)\\ &+ \gamma\Psi(\mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) - \mathbf{I}))\mathbf{B}^{-1}(t).\end{aligned} \tag{5}$$

The traditional ZND model (4) for solving the DLCME is thus obtained.

3 Improved neural dynamics

In this section, a non-convex activation function and a residual-based adaptive coefficient are proposed to release the limitations of the convex activation function and improve the adaptation of the residual error to decrease with time in comparison with the traditional GND model (2) and the traditional ZND model (4).

3.1 Non-convex activation function

Assuming $P_\Pi(\mathbf{O}) = \arg\min_{G\in\Pi}\|\mathbf{G} - \mathbf{O}\|_{\mathrm{F}}$ with $0 \in \Pi$, where $P_\Pi(\mathbf{O})$ is the projection of $\mathbf{O}$ onto Π. Hence, in Eq. (4), it can replace $\Psi(\cdot)$ in it with $P_\Pi(\cdot)$ and then obtain:

$$\dot{\mathbf{E}}(t) = -\gamma P_\Pi(\mathbf{E}(t)). \tag{6}$$

Furthermore, three construction methods for non-convex activation functions are given as follows:

(1) Ball situation with saturation activation function: $\Pi = \{\mathbf{O} \in \mathbb{R}^{n\times n}, \|\mathbf{O}\|_{\mathrm{F}} \le \sigma\}$, where $\sigma > 0$. For this case, $P_\Pi(\mathbf{O}) = \begin{cases}\mathbf{O}, & \|\mathbf{O}\|_{\mathrm{F}} \le \sigma,\\ \sigma\dfrac{\mathbf{O}}{\|\mathbf{O}\|_{\mathrm{F}}}, & \|\mathbf{O}\|_{\mathrm{F}} > \sigma.\end{cases}$

(2) Bound situation with saturation activation function: $\Pi = \{\mathbf{O} \in \mathbb{R}^{n\times n}, \ \delta^- \le \mathbf{O} \le \delta^+\}$, where $\delta^- < 0$ and $\delta^+ > 0$. For this case, $P_\Pi(\mathbf{O}_{ij}) = \begin{cases}\delta_{ij}^+, & \mathbf{O}_{ij} > \delta_{ij}^+,\\ \mathbf{O}_{ij}, & \delta_{ij}^- \le \mathbf{O}_{ij} \le \delta_{ij}^+,\\ \delta_{ij}^-, & \mathbf{O}_{ij} < \delta_{ij}^-.\end{cases}$

(3) Non-convex situation activation function: $\Pi = \{\mathbf{O} \in \mathbb{R}^{n\times n}, \ -c_1 \le \mathbf{O}_{ij} \le c_1 \ \text{ or } \ \mathbf{O}_{ij} = c_2 \ \text{ or } \ \mathbf{O}_{ij} = c_3\}$, where c_1, c_2 and c_3 are constants and

$c_2 > c_1 > 0 > -c_1 > c_3$ at the same time. For this case, $P_\Pi(\mathbf{O}_{ij}) = \begin{cases} c_2, & \mathbf{O}_{ij} > c_1, \\ \mathbf{O}_{ij}, & -c_1 \le \mathbf{O}_{ij} \le c_1, \\ c_3, & \mathbf{O}_{ij} < -c_1. \end{cases}$

Remark 1 Significantly, previous activation functions have a restrictive premise that requires the function to be a continuously derivable and monotonically increasing odd function. The contribution of the non-convex activation function is pioneering in that it allows the activation function to be non-continuous, non-derivable, non-monotonic, and non-odd, relaxing the high constraint on the activation function. The proposal of such an activation function that allows non-derivable non-continuous functions is of great significance because control systems such as robots, cars, or rockets are often non-continuous and non-derivable, and it brings our proposed theoretical algorithms closer to practical applications, so it necessary to use non-convex activation functions to satisfy such an objective physical constraint.

3.2 Residual-based adaptive coefficient

In [36], the researchers pointed that an appropriate coefficient can reduce the convergence time of the traditional neural dynamics. Based on this idea, a residual-based adaptive coefficient function (RBACF) is proposed in this paper, and the IZND activated by the RBACF can be expressed as:

$$\dot{\mathbf{E}}(t) = -\Xi\left(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|\right)P_\Pi(\mathbf{E}(t)), \tag{7}$$

where $\Xi(\cdot)$ denotes the residual-based adaptive coefficient function. An example of the RBACF is shown as

$$\boldsymbol{\Xi}\left(\left|\frac{\partial\|\mathbf{E(t)}\|_{\mathrm{F}}}{\partial \mathbf{t}}\right|\right) = \begin{cases} 10^2\left(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|\right) + 2, & 0 < t < 0.005, \\ 10^6\left(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|\right) + 12, & 0.005 \le t \le 1, \\ 10^p\left(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|\right) + 1, & t > 1, \end{cases} \tag{8}$$

where $p > 0$ is a parameter.

For the comparative purpose, some examples of time-varying adaptive coefficient functions (TVCFs) presented in [22] are listed below.

(1) Time-varying coefficient function 1 (TVCF1):
$\gamma(t) = t^p + p, \quad p > 0.$

(2) Time-varying coefficient function 2 (TVCF2):
$\gamma(t) = p^t + p, \quad p > 0.$

(3) Time-varying coefficient function 3 (TVCF3):
$\gamma(t) = pe^t, \quad p > 0.$

(4) Time-varying coefficient function 4 (TVCF4):
$\gamma(t) = \begin{cases} t^p + p, & 0 < p \le 1, \\ p^t + 2pt + p, & p > 1. \end{cases}$

It is worth mentioning that among the above TVCFs and the proposed RBACF (8), the difference is that the proposed RBACF (8) is related to the current residual error and the TVCFs are only related to the time. This key difference improves the convergence of the proposed IZND model.

Remark 2 Since the initial phase of the problem is a random and disordered state, the bias of the residuals generated by the solution system with respect to time will be large at this time, so only a small gain coefficient is enough to accelerate the convergence of the model. After the initial rapid decline, the computational solution approaches the theoretical solution, but still does not fully converge to the theoretical solution. And after the computational solution of the model converges to the theoretical solution, that is, the residual of the system is small enough, the solution system only needs to make a small adjustment at each moment to keep convergence. Therefore, in the third segment of the segmentation function, we design a gain coefficient that is flexibly adjusted according to the actual problem to keep the model converged with minimal consumption of computational resources.

Remark 3 Although the IZND model (7) is proposed for dynamic linear equations, a review of the existing work shows that Zhang et al. [37], Li et al. [38], and Ma et al. [39] have fully demonstrated the effectiveness of the traditional ZND model when used to solve nonlinear equations and have made many contributions to the field by proposing an optimized ZND model for better solving nonlinear equations. Therefore, the proposed IZND model (7) can be extended to the field of nonlinear equation by the methods in these references, which means that the proposed IZND model (7) is equally applicable in the field of nonlinear equations. In addition, the proposed IZND model (7) incorporates a segmented adaptive coefficient and a non-convex activation function capable of lifting convex constraints, which enables the IZND model (7) to improve its performance in solving nonlinear equations over the traditional ZND model and to have better adaptability when applied to practical.

4 Theoretical analysis

In this section, the global convergence and robustness of the proposed IZND model (7) are analyzed.

4.1 Global convergence

To demonstrate the global convergence of the IZND model (7), the following theorem is proposed to analyze its stability performance based on the Lyapunov stability theory.

Theorem 1 *The computational solution of the IZND model (7) will globally converge to the theoretical solution of the DLCME.*

Proof of Theorem 1 (where $\Xi(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|)$ is written as Ξ for convenience) as

$$\dot{\mathbf{E}}(t) = -\Xi(P_{\Pi}(\mathbf{E}_r(t)) + iP_{\Pi}(\mathbf{E}_i(t))), \tag{9}$$

in which the matrices $\mathbf{E}_r(t)$ and $\mathbf{E}_i(t)$ are the real and imaginary parts of error function $\mathbf{E}(t)$, respectively. In addition, each sub-part of the IZND model (7) can be represented as:

$$\dot{\epsilon}_{jk}(t) = -\Xi(P_{\Pi}(\epsilon_{jk,r}(t)) + iP_{\Pi}(\epsilon_{jk,i}(t))), \tag{10}$$

where $\epsilon_{jk,r}(t)$ and $\epsilon_{jk,i}(t)$ are the sub-parts of $\mathbf{E}_r(t)$ and $\mathbf{E}_i(t)$, respectively. The subscript jk represents the element of the jth row and kth column of the corresponding matrix. Whereafter, consider a Lyapunov candidate function as

$$l_r(t) = \epsilon_j k, r^2(t)/2. \tag{11}$$

Whenever $\epsilon_j k, r(t) \neq 0$, $l_r(t) > 0$. Hence, the Lyapunov candidate function $l_r(t) = \epsilon_j k, r^2(t)/2$ is positive definite. Next, taking derivatives on both sides of (11) can clearly get:

$$\dot{l}_r(t) = -\epsilon_{jk,r}(t)\Xi P_{\Pi}(\epsilon_{jk,r}(t)). \tag{12}$$

Whenever $\epsilon_{jk,r}(t) \neq 0$, it has $\dot{l}_r(t) < 0$. Therefore, $\dot{l}_r(t)$ is negative definite. Since $l_r(t)$ is positive definite and $\dot{l}_r(t) < 0$ is negative definite, the IZND model globally converges, the convergence proof of the real component $\epsilon_{jk,r}(t)$ is accomplished. Similarly, the global convergence can be proved by the approach mentioned earlier for the imaginary component $\epsilon_{jk,i}(t)$. In short, the error synthesized by IZND (7) globally converges to zero; the proof is completed. □

4.2 Robustness under constant noise

For the performance of robustness of the IZND model (7) under constant noise perturbation, this paper proposes the following theorem.

Theorem 2 *The residual error $\|\mathbf{E}(t)\|_{\mathrm{F}}$ of the IZND model (7) perturbed by constant noise η globally converges to the value $\sqrt{P_{\Pi}^{-1}(\eta_{jk,r}/\Xi)^2 + P_{\Pi}^{-1}(\eta_{jk,i}/\Xi)^2}$, which is related to the noise level η, and the adaptive coefficient $\Xi(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|)$ is abbreviated as Ξ.*

Proof of Theorem 2 The IZND model (7) under a constant noise η perturbation can be expressed as:

$$\dot{\mathbf{E}}(t) = -\Xi P_{\Pi}(\mathbf{E}(t)) + \eta. \tag{13}$$

Its sub-part can be simplified as:

$$\dot{\epsilon}_{jk}(t) = -\Xi P_{\Pi}(\epsilon_{jk}(t)) + \eta_{jk}, \tag{14}$$

where ϵ_{jk} is the jkth sub-part of $\mathbf{E}(t)$. Continuing the derivation, $\dot{\epsilon}_j k(t)$ can be rewritten as:

$$\begin{aligned}\dot{\epsilon}_{jk}(t) = {} & \eta_{jk,r} + i\eta_{jk,i} \\ & - \Xi(P_{\Pi}(\epsilon_{jk,r}(t)) + iP_{\Pi}(\epsilon_{jk,i}(t))),\end{aligned} \tag{15}$$

in which $\epsilon_{jk,r}(t)$ and $\epsilon_{jk,i}(t)$ represent the real and imaginary components of $\dot{\epsilon}_{jk}(t)$, respectively. Moreover, $\eta = \eta_{jk,r} + i\eta_{jk,i}$. On the basis of the above derivation, designing a Lyapunov candidate function as

$$Y_{jk}(t) = \epsilon_{jk,r}^2(t)/2. \tag{16}$$

Combining the derivative of $Y_{jk}(t)$ in (16) with Eq. (14), which readily obtain

$$\dot{Y}_{jk}(t) = -\epsilon_{jk,r}(t)(\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r}). \tag{17}$$

It is obvious that the numerical value of $\dot{\mathbf{Y}}_{ij}(t)$ would be influenced with the notation of $\epsilon_{jk,r}(t)$ and $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r}$. Therefore, discussing it in detail by dividing it into the following cases.

(1) When $\epsilon_{jk,r}(t) > 0$:

- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} > 0$, it can readily observe that $Y_{jk}(t) > 0$ and $\dot{Y}_{jk}(t) < 0$. Simultaneously, $\Xi P_{\Pi}(\epsilon_{jk,r}(t))$ will continue to decreasing until $\epsilon_{jk,r}(t)(\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r}) = 0$.

Here, $Y_{jk}(t) > 0$ is positive definite but $\dot{Y}_{jk}(t) < 0$ is negative definite. Therefore, based on the Lyapunov stability theory, $\epsilon_{jk,r}(t)$ possesses global convergence, which means that $\lim_{t\to+\infty} -(\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r})\epsilon_{jk,r}(t) = 0$.

- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} = 0$, since $Y_{jk}(t) > 0$ and $\dot{Y}_{jk}(t) = 0$, $\epsilon_{jk,r}(t)(\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r}) = 0$. Further derivation yields $\epsilon_{jk,r}(t) = P_{\Pi}^{-1}(\eta_{uij}/\Xi)$, where $P_{\Pi}^{-1}(\cdot)$ represents the inverse function of $P_{\Pi}(\cdot)$. Therefore, $\epsilon_{jk,r}(t)$ is equivalent for a particular value when $t \to +\infty$.
- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} < 0$, $Y_{jk}(t) > 0$ and $\dot{Y}_{jk}(t) > 0$ as well, the IZND model (7) is undoubtedly divergent. But $\dot{Y}_{jk}(t)$ will eventually satisfy $\epsilon_{jk,r}(t)(\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r}) = 0$, which means the system will tend to be stabilized. Under such a sub-case, $\epsilon_{jk,r}(t)$ would ultimately converge to a stationary value of $P_{\Pi}^{-1}(\eta_{jk,r}/\Xi)$.

(2) When $\epsilon_{jk,r}(t) = 0$:

- In this case $P_{\Pi}(\epsilon_{jk,r}(t)) = 0$, substituting it into Eq. (17), it get $\dot{\epsilon}_{jk,r}(t) = \eta_{jk,r}$. Thus, it can easily find that the change of ϵ_{ij} depends on the sign of $\eta_{jk,r}$. So, $P_{\Pi}(\epsilon_{jk,r}(t))$ in here is a temporary state that will be transformed into case (1) or case (3) in the next moment.

(3) When $\epsilon_{jk,r}(t) < 0$:

- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} > 0$, $\dot{Y}_{jk}(t) > 0$. Since $\epsilon_{jk,r}(t) < 0$, $P_{\Pi}(\epsilon_{jk,r}(t))$ increases as $\epsilon_{jk,r}(t)$ increases. Importantly, when $P_{\Pi}(\epsilon_{jk,r}(t)) \leq \epsilon_{jk,r}(t)$, even if $\epsilon_{jk,r}(t) > 0$, it is still the case that the residual error is upper bounded by $P_{\Pi}^{-1}(\eta_{jk,r}/\Xi)$.
- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} = 0$, it is evident that $\dot{Y}_{jk}(t) = 0$. Clearly, $\epsilon_{jk,r}(t)$ globally converges to a certain value $P_{\Pi}^{-1}(\eta_{jk,r}/\Xi)$.
- For the case $\Xi P_{\Pi}(\epsilon_{jk,r}(t)) - \eta_{jk,r} < 0$, in this case it is the same as the first sub-case in case (1), so it is simple to obtain that $\epsilon_{jk,r}(t)$ is global convergent.

Based on the above, it follows in the same way that the imaginary part $\epsilon_j k, i(t)$ of $\epsilon_j k(t)$ can globally converge to $P_{\Pi}^{-1}(\eta_j k, i/\Xi)$. Finally, it can make the conclusion that the residual error $\|\mathbf{E}(t)\|_{\mathrm{F}}$ globally converges to a particular value $\sqrt{P_{\Pi}^{-1}(\eta_{jk,r}/\Xi)^2 + P_{\Pi}^{-1}(\eta_{jk,i}/\Xi)^2}$. The proof is completed. □

4.3 Robustness under time-varying noise

Theorem 3 *For the IZND model (7) under bounded dynamic noise $\eta(t)$ perturbation, its residual error $\|\mathbf{E}(t)\|_{\mathrm{F}}$ will ultimately converge to a particular domain $\lim_{t\to+\infty} \sqrt{\sum_{j=1,k=1}^{nm} \varrho^2}/c_2$. In the above parameters, $\varrho = \max\|\eta(t)\|$ and jk represents the jth row and kth column of the matrix.*

Proof of Theorem 3 Hereby, to simplify the proof process, replacing the non-convex activation function $P_{\Pi}(\cdot)$ by the linear activation function $\Psi_{li}(\cdot)$. Thus, it obtains $\dot{E}(t) = -\Xi\Phi(\mathbf{E}(t))$ according to Eq. (7), whereupon its expression under dynamic noise $\eta(t)$ perturbation can be described as $\dot{E}(t) = -\Xi\Psi_{li}(\mathbf{E}(t)) + \eta(t)$, while its subcomponent is derived as:

$$\dot{\epsilon}_{jk}(t) = -\Xi\Psi_{li}(\epsilon_{jk}(t)) + \eta_{jk}(t), \tag{18}$$

note that here $j \in [1, n]$ and $k \in [1, m]$. Further deriving the general solution of its first-order differential equation and then substituting it into Eq. (18) to obtain:

$$\begin{aligned}\epsilon_{jk}(t) &= \epsilon_{jk}(0)e^{-(c_2 t + c_1\int_0^t |\frac{\partial\|\mathbf{E}(\tau)\|_{\mathrm{F}}}{\partial\tau}|\mathrm{d}\tau)} \\ &\quad + e^{-(c_2 t + c_1\int_0^t |\frac{\partial\|\mathbf{E}(\tau)\|_{\mathrm{F}}}{\partial\tau}|\mathrm{d}\tau)} \\ &\quad \times \int_0^t e^{c_2\tau + c_1\int_0^t |\frac{\partial\|\mathbf{E}(\tau)\|_{\mathrm{F}}}{\partial\tau}|\mathrm{d}\tau}|\eta_{jk}(\tau)|\mathrm{d}\tau.\end{aligned}$$

We simplify and rearrange the above equation as

$$\epsilon_{jk}(t) = \epsilon_{jk}(0)e^{-\Delta} + e^{-\Delta}\int_0^t e^{\Delta}|\eta_{jk}(\tau)|\mathrm{d}\tau, \tag{19}$$

where the parameter $\Delta = c_2 t + c_1\int_0^t |\frac{\partial\|\mathbf{E}(\tau)\|_{\mathrm{F}}}{\partial\tau}|\mathrm{d}\tau > 0$. Based on the triangle inequality, the equation can further be derived as:

$$|\epsilon_{jk}(0)| \leq |\epsilon_{jk}(0)e^{-\Delta}| + |e^{-\Delta}\int_0^t e^{\Delta}|\eta_{jk}(\tau)|\mathrm{d}\tau|. \tag{20}$$

Letting $\eta_{jk}(\tau)$ reach its maximum value in the domain of definition while $t \to +\infty$ and considering the L'Hubida's law yields:

$$|\epsilon_{jk}(t)| \leq \frac{\max|\eta_{jk}(\delta)|}{|c_2 + c_1|\frac{\partial\|\mathbf{E}(\mathbf{t})\|_{\mathrm{F}}}{\partial t}||}, \quad 0 \leq \delta \leq t.$$

On the basis of the above derivation, $\epsilon(t)$ fulfills the condition:

$$\lim_{t\to+\infty}\|\epsilon(t)\|_2 \leq \frac{\|\varrho\|_2}{c_2}. \tag{21}$$

Deriving to this point, we can conclude from Eq. (21) that the upper bound of the residual error $\epsilon(t)$ is inversely proportional to the adaptive system c_2 and positively proportional to the maximum value ϱ of the noise $\eta_{jk}(t)$. $\square$

4.4 Time complexity

To demonstrate for the feasibility of the proposed IZND model (7) from an implementation and computational point of view, a theoretical analysis of the time complexity of the model containing smoothly dynamic matrices $\mathbf{A} \in \mathbb{C}^{n\times n}$, $\mathbf{B} \in \mathbb{C}^{n\times n}$ and identity matrix $\mathbf{I} \in \mathbb{C}^{n\times n}$ is presented.

- Time complexity of obtaining $\|\mathbf{E}(t)\|_{\mathrm{F}}$: $O(n^3)$. Time complexity of obtaining $\mathbf{E}(t)$ is $O(2n^3 + 2n^2)$; time complexity of obtaining $\|\cdot\|_{\mathrm{F}}$ is $O(n^2)$. The total time complexity of this sub-part is $O(2n^3 + 3n^2)$, since the complexity in each sub-part is governed by the highest order of complexity, the total time complexity here is $O(n^3)$.
- Time complexity of obtaining $\Xi\left(\left|\frac{\partial\|\mathbf{E}(t)\|_{\mathrm{F}}}{\partial t}\right|\right)$: $O(n^3)$. Time complexity of obtaining $\|\mathbf{E}(t)\|_{\mathrm{F}}$ is $O(2n^3 + 3n^2)$, time complexity of obtaining $\frac{\partial\cdot}{\partial t}$ is $O(1)$, time complexity of obtaining $\Xi(\cdot)$ is $O(1)$. The total time complexity of this sub-part is $O(2n^3 + 3n^2 + 2)$, since the complexity in each sub-part is governed by the highest order of complexity, the total time complexity here is $O(n^3)$.
- Time complexity of obtaining $P_\Pi(\mathbf{E}(t))$: $O(n^3)$. Time complexity of obtaining $\|\mathbf{E}(t)\|_{\mathrm{F}}$ is $O(2n^3 + 3n^2)$; time complexity of obtaining $P_\Pi(\cdot)$ is $O(n^2)$. The total time complexity of this sub-part is $O(2n^3 + 4n^2)$, since the complexity in each sub-part is governed by the highest order of complexity, the total time complexity here is $O(n^3)$.

In summary, the total time complexity (and corresponding operational complexity) of the proposed IZND model (7) is $O(n^3)$.

5 Simulative verifications

Throughout this section, two examples have proposed to demonstrate the effectiveness and superiority of the IZND model (7) in solving the DLCME.

5.1 Numerical simulation in low dimension

In this example, defining dynamic complex-valued matrices $\mathbf{A} \in \mathbb{C}^{2\times 2}$ and $\mathbf{B} \in \mathbb{C}^{2\times 2}$ as.

$$\mathbf{A}(t) = \begin{bmatrix} \exp(it) & -i\exp(it) \\ -i\exp(it) & \exp(it) \end{bmatrix},$$

$$\mathbf{B}(t) = \begin{bmatrix} \exp(2it) & -2i\exp(-4it) \\ -2i\exp(4it) & \exp(-2it) \end{bmatrix}.$$

Therefore, through calculation, it gets the theoretical solution $\mathbf{X}^*(t)$ of $\mathbf{A}(t)\mathbf{X}(t)\mathbf{B}(t) = \mathbf{I}$ as

$$\mathbf{X}^*(t) = \begin{bmatrix} 0.1\exp(-3it) - 0.2\exp(3it) & 0.1i\exp(it) + 0.2i\exp(-5it) \\ 0.1i\exp(-it) + 0.2i\exp(5it) & 0.1\exp(3it) - 0.2\exp(-3it) \end{bmatrix}. \tag{22}$$

Substituting the non-convex activation function into IZND model (7), it can draw a image Fig. 1 that compares the computed solution with the theoretical solution of the DLCME. According to Fig. 1, the computed solution overlaps with the theoretical solution of the DLCME in a short time from a stochastically produced original value. In addition, in Fig. 2, the superiority of using the IZND model (7) activated by the RBACF (8) compared to the traditional ZND model (4) and the traditional GND model (2) is demonstrated. Also, it is proved that the IZND model (7) outperforms the traditional ZND model (4) and the traditional GND model (2) in terms of convergence speed and convergence accuracy. In addition, through Fig. 3, it can clearly see the advantages of the adaptive coefficient based on residual error $\|\mathbf{E}(t)\|_{\mathrm{F}}$ in our proposed model: for the TVCFs mentioned in Sect. 3, the adaptive coefficient has great advantages in convergence accuracy and convergence speed. Furthermore, as can be seen in Figs. 4 and 5, the IZND model (7) converges to 10^{-2} and 10^{-3}, respectively, demonstrating excellent noise rejection performance. With the addition of constant noise $\eta = 10$ and bounded dynamic noise $\eta = 10t$, respectively, the model converges to zero within a short period of time, and its convergence is much better than that of the conventional ZND model (4) and the conventional GND model (2). More convincingly,

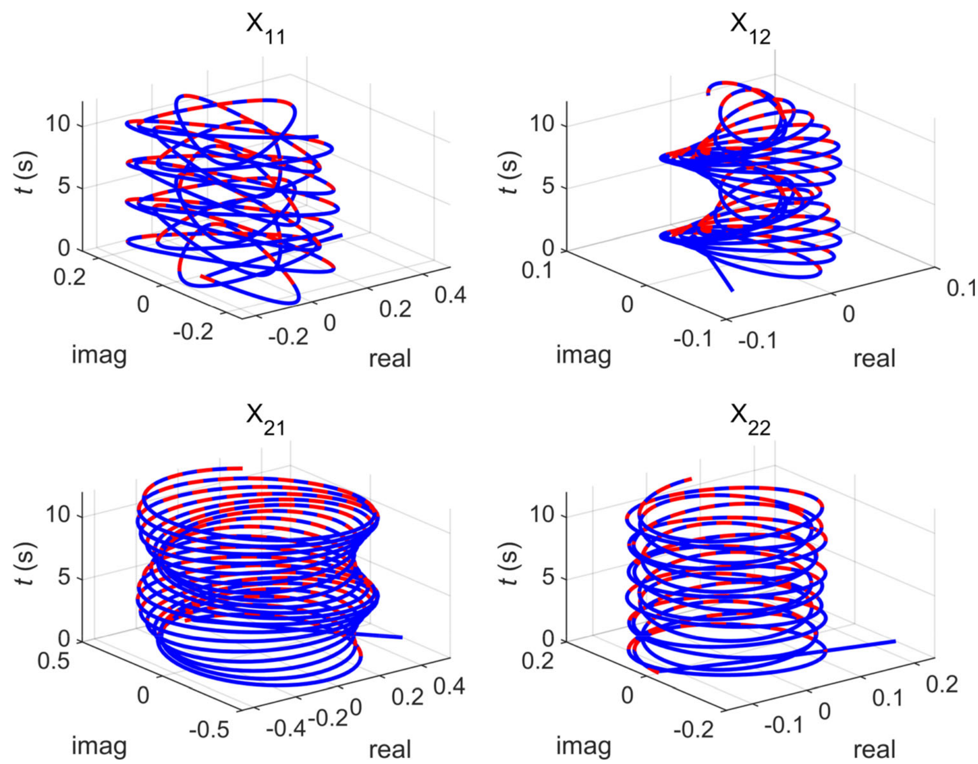
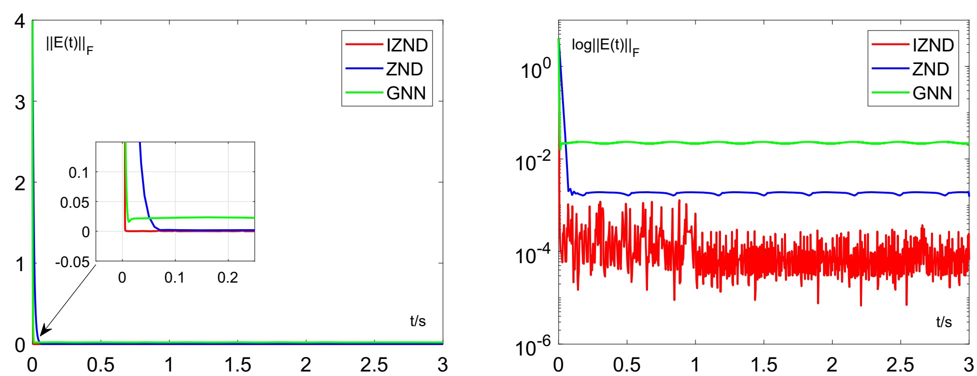
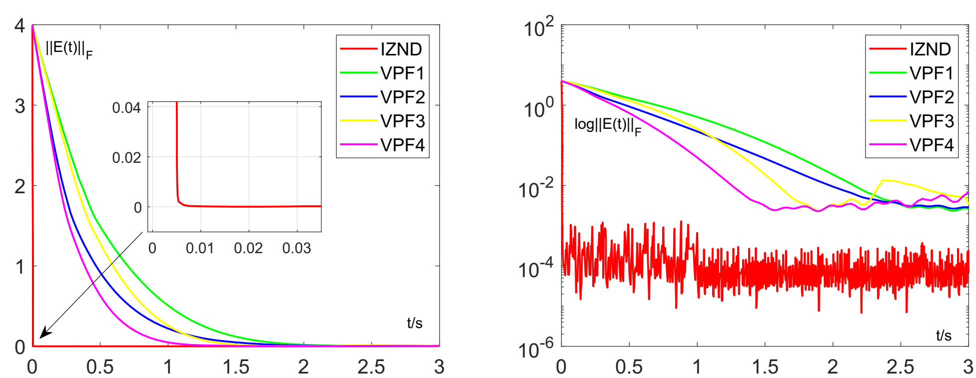


Fig. 1 Trajectories of the theoretical (red dashed line) and computed (blue solid line) solutions obtained for the IZND model (7) with c1 = 1, c2 = 2, and c3 = 2 in solving the low-dimensional DLCME (Color figure online)



(a) (b)

Fig. 2 **a** The comparison among the IZND model (7) activated by a non-convex activation function with c1 = 1, c2 = 2, c3 = −2 (red solid line), the conventional ZND model (4) with $\gamma = 100$ (blue solid line) and the conventional GND model (2) with $\beta = 100$ (green solid line) activated by a linear activation function. **b** The logarithmic plot of **a** (Color figure online)



(a) (b)

Fig. 3 **a** Comparison among the convergence of the IZND models (7) (red solid line) with other ZND models (green, blue, yellow, and purple solid lines) using four different time-varying coefficient functions, all of these models are activated by non-convex activation functions with c1 = 1, c2 = 2, and c3 = −2. **b** The logarithmic plot of **a** (Color figure online)

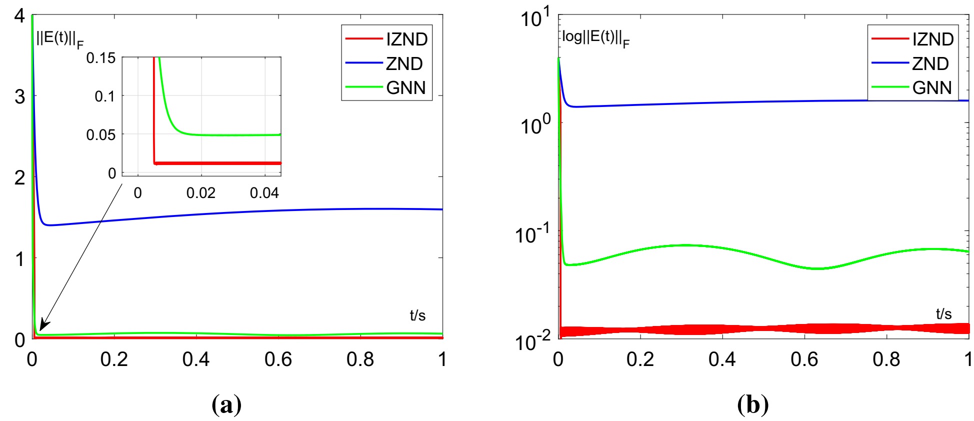


Fig. 4 **a** Comparison among the convergence of the IZND model (7) (red solid line), the conventional ZND model (4) with $\gamma = 100$ (blue solid line) and the conventional GND model (2) with $\beta = 100$ (green solid line) under constant noise $\eta = 10$ perturbation. **b** The logarithmic plot of **a** (Color figure online)

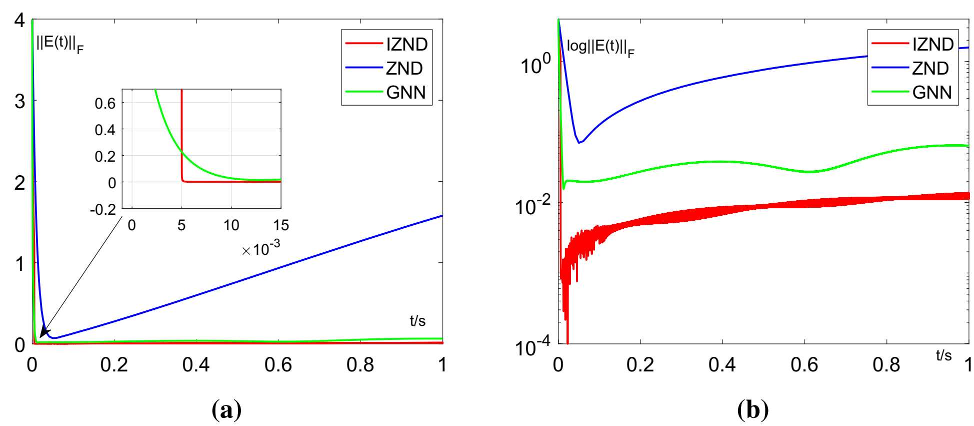


Fig. 5 **a** Comparison among the convergence of the IZND model (7) (red solid line), the conventional ZND model (4) with $\gamma = 100$ (blue solid line) and the conventional GND model (2) with $\beta = 100$ (green solid line) under constant noise $\eta = 10t$ perturbation. **b** The logarithmic plot of **a** (Color figure online)

Table 1 The characteristics of the average steady-state residual error (ASSRE) and the maximum steady-state residual error (MSSRE) in solving the DLCME problem (1) are compared between the traditional matrix inversion method, the GND model (2), the ZND model (4), and the proposed IZND model (7)

Model	Parameter	ASSRE		MSSRE	
		With CN	With TVN	With CN	With TVN
TMI	None	1.415×10^{2}	2.088×10^{2}	1.611×10^{2}	4.015×10^{2}
GND (2)	$\beta = 20$	7.022×10^{0}	3.488×10^{0}	7.278×10^{0}	6.955×10^{0}
	$\beta = 100$	1.553×10^{0}	8.588×10^{-1}	1.603×10^{0}	1.580×10^{0}
ZND (4)	$\gamma = 20$	3.039×10^{-1}	1.935×10^{-1}	3.660×10^{-1}	3.225×10^{-1}
	$\gamma = 100$	6.100×10^{-2}	3.923×10^{-2}	7.337×10^{-2}	6.463×10^{-2}
IZND (7)	$p = 1.5$	$\mathbf{1.240 \times 10^{-2}}$	$\mathbf{9.566 \times 10^{-3}}$	$\mathbf{1.363 \times 10^{-2}}$	$\mathbf{1.362 \times 10^{-2}}$

The optimal values for each condition are shown in bold

★TMI represents the method of traditional matrix inversion

CN and TVN indicate constant noise 10 and time-varying noise 10t, respectively

from Table 1, this paper compares the average steady-state error and maximum steady-state error of the traditional matrix inversion method, the GND model (2), the ZND model (4) and the proposed IZND model (7), fully demonstrating the superiority of the IZND model (7). Also, as a high-performance algorithm, the IZND model (7) has a time complexity of about 10^{-4} in the absence of noise, which is also far superior to the GND (2) and ZND (4) models.

5.2 Numerical simulation in high dimension

Set dynamic complex-valued matrices $\mathbf{A} \in \mathbb{C}^{4\times4}$ and $\mathbf{B} \in \mathbb{C}^{4\times4}$, which are as follows:

$$\mathbf{A}(t)=\begin{bmatrix} sin(t)+i & cos(t)+sin(2t)i & sin(t)-cos(t)i & 2+cos(t)i \\ 2+cos(t)i & cos(t)-sin(t)i & 1+2cos(2t)i & cos(3t)+i \\ sin(2t)+3i & cos(2t)+sin(t)i & sin(t)-3i & sin(t)+2cos(t)i \\ sin(t)-2i & 2+sin(3t)i & sin(2t)-3i & 2+cos(t)-sin(t)i \end{bmatrix},$$

$$\mathbf{B}(t)=\begin{bmatrix} t & 1+t & 1+t & 1+t \\ 1+t & t & 1+t & 1+t \\ 1+t & 1+t & t & 1+t \\ 1+t & 1+t & 1+t & t \end{bmatrix}.$$

Figure 6 shows the simulated solution calculated by the IZND model (7) and the theoretical solution of the dynamic complex-valued matrix in the case of a high-dimensional matrix where it can be known that the model has excellent performance.

5.3 Example simulation of acoustic source localization

This subsection puts forward a simulation example of acoustic source localization built on the time difference of arrival (TDOA) method. By exploiting the IZND model (7) proposed above, this paper achieves real-time high-precision localization of moving targets.

According to the research done by Jin and Yan et al [7], the TDOA method can be applied to the field of acoustic source localization, and the location of the acoustic source can be easily obtained by calculating the time difference between the arrival of the acoustic source signal at different locations of the receiver. In the real world, this method can be implemented not only in the field of acoustic source localization, but also in the field of mobile communication [40] and indoor localization [41].

In this simulation study, for the sake of simplicity, we only study and validate the two-dimensional acoustic source localization problem, which can be readily expanded to the three-dimensional acoustic source localization situation. Further, the theoretical trajectory $\mathbf{E}^*(t)$ of the acoustic source is defined as

$$\mathbf{E}^*(t)=\begin{bmatrix} 25+4cos(1.44t)+10cos(0.96t) \\ 25+4sin(1.44t)-6sin(0.96t) \end{bmatrix}\in\mathbb{R}^2.$$

Simplistically, we assign the coordinates of the n acoustic sensors used to receive acoustic signals as

$$\mathbf{R}^*(t)=\begin{bmatrix} 25+4cos(1.44t_1)+10cos(0.96t_1) & \cdots & \cdots & 25+4cos(1.44t_n)+10cos(0.96t_n) \\ 25+4sin(1.44t_1)-6sin(0.96t_1) & \cdots & \cdots & 25+4sin(1.44t_n)-6sin(0.96t_n) \end{bmatrix}.$$

Theoretically, the acoustic source moves along a pentagram trajectory with time, and acoustic sensor locations are boundedly random. In accordance with the theory of the TDOA method, the number of acoustic sensors should be larger than or equal to three because the studied sound source localization is on a two-dimensional scale. Here, we declare the time difference between the arrival time of each subsensor and the arrival time of the main sensor as τ_i and the length of the coordinates of each sensor in geometric space $d_i^2=x_i^2+y_i^2+z_i^2$. Thus, the distance difference from each subsensor to the main sensor is the product of the time difference τ_i between the arrival time of each subsensor and the arrival time of the main sensor and the speed of sound propagation v, i.e., $\Delta r_i=v\tau_i$. In this study, only four acoustic sensors are considered for the sake of simplicity and accuracy, and the equation for n sensors is

$$\begin{bmatrix} S_{x_1}-S_{x_0} & S_{y_1}-S_{y_0} \\ \vdots & \vdots \\ S_{x_n}-S_{x_0} & S_{y_n}-S_{y_0} \end{bmatrix}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}= r_0\begin{bmatrix} -\Delta r_1 \\ \vdots \\ -\Delta r_n \end{bmatrix}+\begin{bmatrix} l_1 \\ \vdots \\ l_n \end{bmatrix}, \tag{23}$$

where $i\in\{1,2,\ldots,n\}$, and $l_i=\frac{d_i^2-d_0^2-\Delta r_i^2}{2}$.

With strong support from the above equation, our simulations yielded images of Fig. 7. Figure 7a shows a trajectory diagram of the launch path versus the target path, where the four acoustic sensors are labeled. The small difference between the launch path and the target path indicates the high accuracy of the IZND model (7). Figure 7b is the residual error in the x-axis and the y-axis when calculating the launch path. The simulation result shows the superior convergence of the IZND model (7).

6 Conclusions

In this paper, a non-convex activation and residual-based adaptive coefficient zeroing neural dynamic model (IZND) was proposed for online solving the dynamic linear complex matrix equation (DLCME). Noticeably, an adaptive coefficient based on residuals is proposed in this paper, and

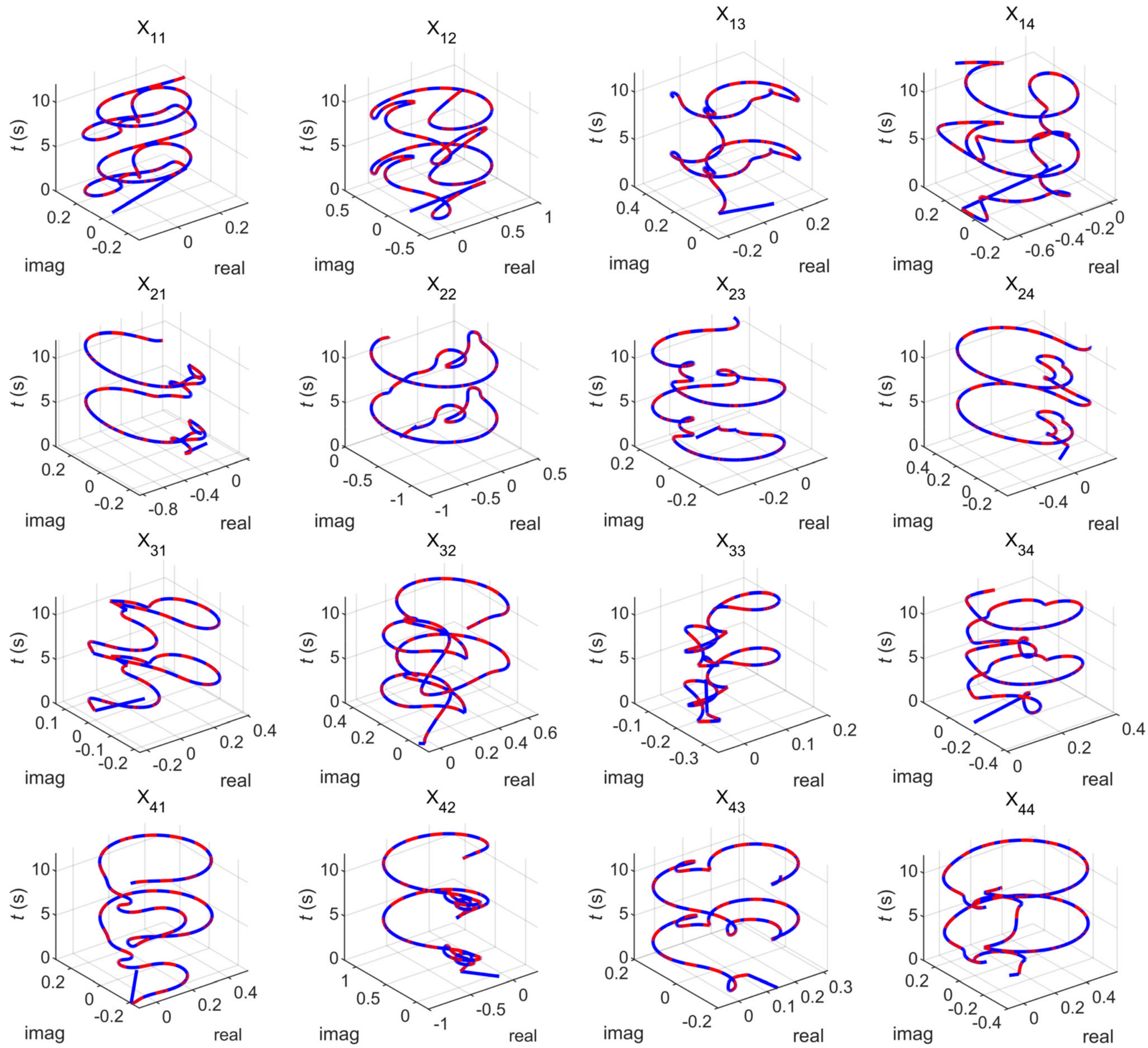


Fig. 6 Trajectories of the theoretical (red dashed line) and computed (blue solid line) solutions obtained for the IZND model (7) with c1 = 1, c2 = 2, and c3 = 2 in solving the high-dimensional DLCME (Color figure online)

a non-convex activation function is applied. This residual-based adaptive coefficient function (RBACF) improves the adaptability of the residual error decreasing with time while greatly improving the convergence speed of the IZND model, and more importantly, it significantly improves the robustness of the model. Secondly, the non-convex activation function is also designed to increase the convergence speed of the IZND model and reduce its upper limit of convergence accuracy, and the most important is that it is not confined to the questions requiring convex activation function. Subsequently, it performs theoretical analyses for the activated complex-valued matrix and then rigorously proves the convergence of the IZND model and the robustness under constant and dynamic noise of the model. In addition, simulation experiments of the DLCME examples in two different dimensions strongly demonstrate the validity of the IZND model. The application of the IZND model in the field of acoustic source localization also demonstrates the generalizability and efficiency of the model. Finally, we will further explore the optimization of the adaptive coefficients in our subsequent work to explore the potential of this segmentation function for accelerating the convergence and enhancing the robustness of the model. We are also exploring the application of such a non-convex saturated IZND model to multi-target dynamic localization, UAV platform algorithms, financial stock

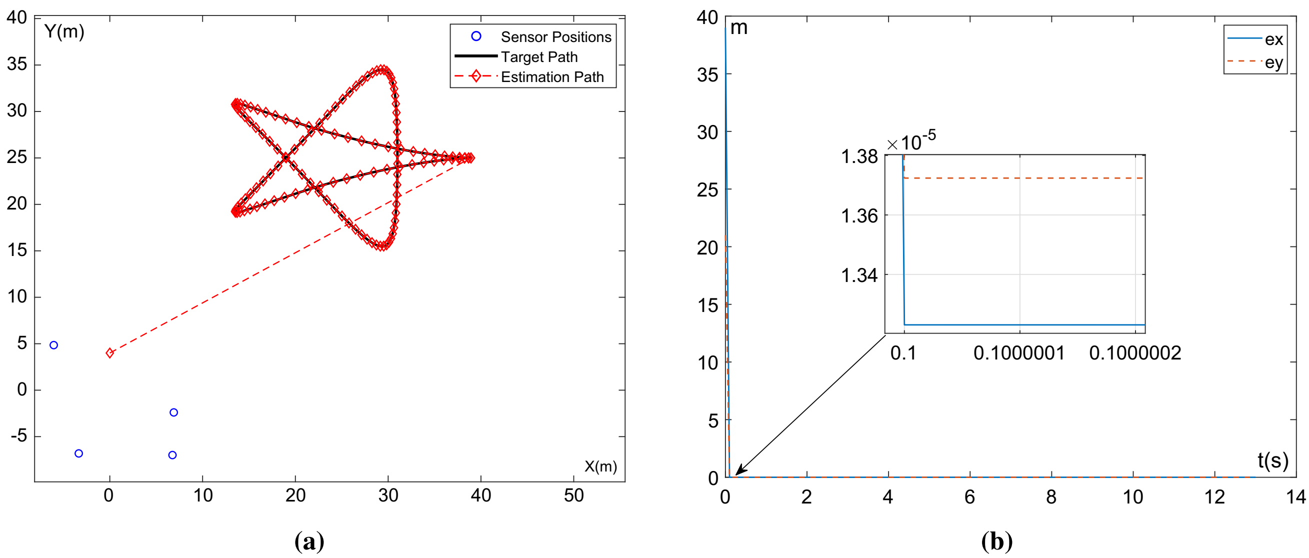


Fig. 7 **a** The trajectory of the IZND model (7) using the TDOA method to estimate the emission path (red with diamond dashed line) and the target path (black solid line) when moving the sound source location, as well as the sensor location distribution. **b** The error in the x-axis (blue solid line) versus the error in the y-axis (red dashed line) during the localization of the moving acoustic source (Color figure online)

market prediction, extreme weather prediction, and other fields.

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Declarations

Conflict of interest We wish to draw the attention of the Editor to the following facts, which may be considered as potential conflicts of interest and to significant financial contributions to this work. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the other of authors listed in the manuscript has been approved by all of us and that the second author prepared the revision information letter and addressed most of the comments. We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property. We understand that the Corresponding Author is the sole contact for the Editorial process (including Editorial Manager and direct communications with the office). He is responsible for communicating with the other authors about progress, submissions of revision and final approval of proof. We confirm that we have provided a current email address, which is accessible by the corresponding author, which has been configured to accept email from the Neural Computing and Applications Editorial Office.

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